

# Credit Card Default Auto-ML Report — lgbm (sub)

 2026-05-28 22:04:26 UTC

모형 lgbm

roc\_auc = 0.7746

BEST(catboost) = 0.7796

Δ vs best = -0.0050

## 1 점수 요약 (본 모형 vs best)

| metric    | lgbm (현재) |        | catboost (BEST) |        | Δ(Test) |
|-----------|-----------|--------|-----------------|--------|---------|
|           | Train     | Test   | Train           | Test   |         |
| roc_auc   | 0.8273    | 0.7746 | 0.8099          | 0.7796 | -0.0050 |
| ks        | 0.5022    | 0.4248 | 0.4660          | 0.4362 | -0.0114 |
| lift      | 3.2530    | 3.1123 | 3.4037          | 3.1274 | -0.0151 |
| accuracy  | 0.8188    | 0.8137 | 0.8308          | 0.8177 | -0.0040 |
| precision | 0.7399    | 0.7061 | 0.7133          | 0.6589 | +0.0472 |
| recall    | 0.2791    | 0.2698 | 0.3931          | 0.3640 | -0.0942 |
| f1        | 0.4054    | 0.3904 | 0.5069          | 0.4689 | -0.0785 |
| pr_auc    | 0.5999    | 0.5430 | 0.6194          | 0.5588 | -0.0158 |

## 2 Feature Importance — lgbm

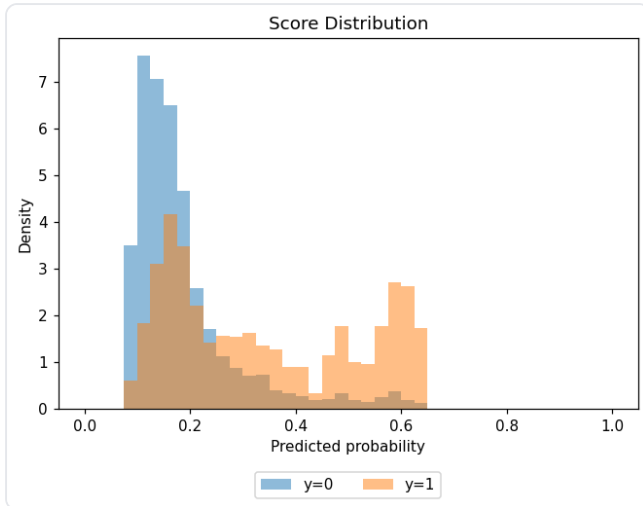
relative · top 19 · 전체 분포는 feature\_importance.csv

| Rank | Feature   | Importance | Rank | Feature   | Importance |
|------|-----------|------------|------|-----------|------------|
| 1    | PAY_0     | 0.4705     | 16   | PAY_5     | 0.0154     |
| 2    | PAY_2     | 0.1389     | 17   | BILL_AMT6 | 0.0149     |
| 3    | LIMIT_BAL | 0.0435     | 18   | PAY_6     | 0.0123     |
| 4    | BILL_AMT1 | 0.0390     | 19   | SEX       | 0.0027     |
| 5    | PAY_AMT3  | 0.0353     |      |           |            |
| 6    | PAY_3     | 0.0349     |      |           |            |
| 7    | PAY_AMT2  | 0.0284     |      |           |            |
| 8    | PAY_AMT1  | 0.0256     |      |           |            |
| 9    | BILL_AMT2 | 0.0243     |      |           |            |
| 10   | PAY_AMT4  | 0.0211     |      |           |            |
| 11   | PAY_4     | 0.0209     |      |           |            |
| 12   | AGE       | 0.0196     |      |           |            |
| 13   | PAY_AMT6  | 0.0184     |      |           |            |
| 14   | PAY_AMT5  | 0.0180     |      |           |            |
| 15   | BILL_AMT5 | 0.0163     |      |           |            |

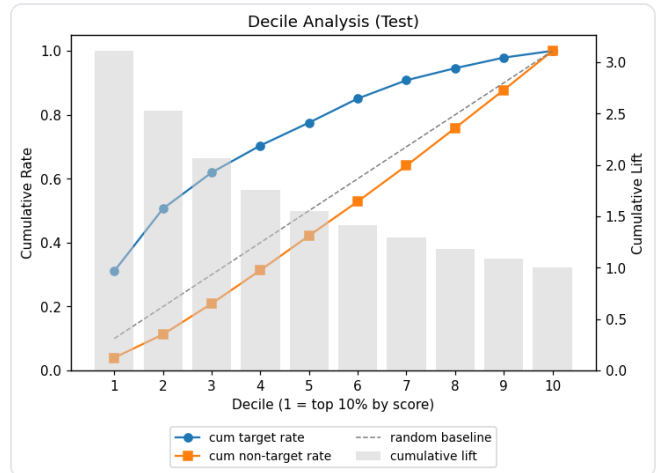
### 3 모델 상세 — lgbm

점수 분포 · 10-분위 차트

Score Distribution by Label



Decile Analysis Chart



| Decile | Score Range   | Count | Target | Non-Target | Cum Target | Cum NonTarget | Cum Count | Cum %  |
|--------|---------------|-------|--------|------------|------------|---------------|-----------|--------|
| 1      | 0.477 ~ 0.630 | 600   | 413    | 187        | 413        | 187           | 600       | 10.0%  |
| 2      | 0.297 ~ 0.477 | 600   | 259    | 341        | 672        | 528           | 1200      | 20.0%  |
| 3      | 0.222 ~ 0.297 | 600   | 150    | 450        | 822        | 978           | 1800      | 30.0%  |
| 4      | 0.188 ~ 0.222 | 600   | 112    | 488        | 934        | 1466          | 2400      | 40.0%  |
| 5      | 0.169 ~ 0.188 | 600   | 95     | 505        | 1029       | 1971          | 3000      | 50.0%  |
| 6      | 0.152 ~ 0.169 | 600   | 100    | 500        | 1129       | 2471          | 3600      | 60.0%  |
| 7      | 0.137 ~ 0.152 | 600   | 76     | 524        | 1205       | 2995          | 4200      | 70.0%  |
| 8      | 0.120 ~ 0.136 | 600   | 50     | 550        | 1255       | 3545          | 4800      | 80.0%  |
| 9      | 0.104 ~ 0.120 | 600   | 44     | 556        | 1299       | 4101          | 5400      | 90.0%  |
| 10     | 0.086 ~ 0.104 | 600   | 28     | 572        | 1327       | 4673          | 6000      | 100.0% |

Confusion Matrix threshold = 0.5

|        | Pred 0 | Pred 1 |
|--------|--------|--------|
| True 0 | 4524   | 149    |
| True 1 | 969    | 358    |

#### 4 변수 선택 (Stability Selection)

base: lasso

부분표본 80회

임계값 0.60

채택 19 / 23

표 노출 23 / 전체 23 (전체 분포는 메인 리포트의 feature\_selection.csv)

| Feature   | Selection Frequency    |        | 채택   |
|-----------|------------------------|--------|------|
| LIMIT_BAL | <div><div></div></div> | 100.0% | ✓ 채택 |
| AGE       | <div><div></div></div> | 100.0% | ✓ 채택 |
| PAY_0     | <div><div></div></div> | 100.0% | ✓ 채택 |
| PAY_2     | <div><div></div></div> | 100.0% | ✓ 채택 |
| PAY_3     | <div><div></div></div> | 100.0% | ✓ 채택 |
| PAY_4     | <div><div></div></div> | 100.0% | ✓ 채택 |
| BILL_AMT1 | <div><div></div></div> | 100.0% | ✓ 채택 |
| PAY_AMT1  | <div><div></div></div> | 100.0% | ✓ 채택 |
| PAY_AMT2  | <div><div></div></div> | 100.0% | ✓ 채택 |
| PAY_AMT3  | <div><div></div></div> | 100.0% | ✓ 채택 |
| PAY_AMT6  | <div><div></div></div> | 100.0% | ✓ 채택 |
| PAY_5     | <div><div></div></div> | 98.8%  | ✓ 채택 |
| BILL_AMT5 | <div><div></div></div> | 98.8%  | ✓ 채택 |
| PAY_6     | <div><div></div></div> | 97.5%  | ✓ 채택 |
| PAY_AMT4  | <div><div></div></div> | 96.2%  | ✓ 채택 |
| PAY_AMT5  | <div><div></div></div> | 95.0%  | ✓ 채택 |
| SEX       | <div><div></div></div> | 91.2%  | ✓ 채택 |
| BILL_AMT6 | <div><div></div></div> | 88.8%  | ✓ 채택 |
| BILL_AMT2 | <div><div></div></div> | 71.2%  | ✓ 채택 |
| BILL_AMT3 | <div><div></div></div> | 56.2%  | 제외   |
| BILL_AMT4 | <div><div></div></div> | 28.7%  | 제외   |
| EDUCATION | <div><div></div></div> | 21.2%  | 제외   |
| MARRIAGE  | <div><div></div></div> | 15.0%  | 제외   |

## 5 학습 설정 + 튜닝 — lgbm

| 키                                  | 값                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tuned                              | <b>YES</b> n_trials = 15 , best_value = 0.7849                                                                                                                                                                                                                                                                                                                          |
| best_iteration<br>(avg over folds) | 154                                                                                                                                                                                                                                                                                                                                                                     |
| params (최종)                        | {'objective': 'binary', 'metric': 'auc', 'verbose': -1, 'n_estimators': 800, 'max_depth': -1, 'learning_rate': 0.021471858797027, 'num_leaves': 85, 'min_data_in_leaf': 55, 'feature_fraction': 0.7779717392187493, 'bagging_fraction': 0.6052561262871233, 'reg_alpha': 0.07115953064092197, 'reg_lambda': 7.860792250906604, 'min_gain_to_split': 0.1471710061864095} |